

Cyclotomic Blow-Ups and Natural Boundaries for the q -Fabius–Rvachev Sinc Product

Spectral dilogarithms, polyharmonic limits, Bell condensation, and formal inverse series

Research report for the ProveIt/Analysis/FabiusFunction project

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Abstract

The dyadic Rvachev up-function has Fourier transform $\prod_{j \geq 1} \text{sinc}_{\text{rad}}(z/2^j)$, and the surrounding repository develops its Fabius, Thue–Morse, q -series, Bell–Bernoulli, inverse, sampling, and Lambert- W structures in considerable depth. The current consolidated q -series frontier explicitly leaves two related tasks open: proving a generic natural boundary in the q -plane and constructing root-of-unity renormalizations. This report resolves a concrete scalar version of both tasks for the complex geometric-uniform transform

$$\Phi(q, z) = \prod_{j \geq 0} \text{sinc}_{\text{rad}}((1 - q)q^j z), \quad |q| < 1.$$

For a primitive m th root of unity ω and $d = \text{ord}_{\text{mult}}(\omega^2) = m / \gcd(m, 2)$, we derive the radial expansion

$$\log \Phi(\omega(1 - \delta), z) = -\frac{\mathcal{A}_\omega(z)}{\delta} + \mathcal{B}_\omega(z) + O(\delta),$$

locally uniformly for $|(1 - \omega)z| < \pi$. The action is an explicit spectral dilogarithm,

$$\mathcal{A}_\omega(z) = \frac{1}{2d^2} \sum_{n \geq 1} \text{Li}_2 \left(\left(\frac{(1 - \omega)z}{\pi n} \right)^{2d} \right).$$

Its sign is never zero for real $0 < |z| < \pi/2$. Because roots of unity are dense, this proves that the unit circle is a natural boundary for $q \mapsto \Phi(q, z)$ throughout that entire frequency interval, strengthening the repository’s generic conjecture to an explicit theorem.

A double scaling $q = \omega(1 - \varepsilon^d)$, $z = \sqrt{\varepsilon} w$ then gives the locally uniform cyclotomic limit

$$\Phi(\omega(1 - \varepsilon^d), \sqrt{\varepsilon} w) \longrightarrow \exp(-A_\omega w^{2d}), \quad A_\omega = \frac{\zeta(2d)(1 - \omega)^{2d}}{2d^2 \pi^{2d}}.$$

We prove the first corrections and an all-orders Bell-polynomial expansion, identify the corresponding polyharmonic heat symbol, show that the reciprocal zero moments and Bernoulli cumulants collapse to a single order $2d$, derive the limiting Gould–Hopper Appell family, and construct formal inverse series both in the scaled frequency and in q itself. A universal analytic-germ theorem shows that the mechanism extends beyond sinc products to q -Pochhammer and other geometric products. High-precision experiments, source code, CSV data, and figures accompany the report.

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1 Scope, status, and contribution boundary

1.1 The audited source landscape

The requested source tree is a living research corpus rather than a single paper. Its current top level contains the primary Fabius–Rvachev exposition, a mathematical glossary, a non-elementarity study, a Lean walkthrough, imported papers, an archive, and a semi-formalized frontier system. During August 2026 many formerly separate frontier drafts were consolidated into thematic volumes. Consequently, a novelty audit based only on old filenames would be misleading: the correct comparison target is the current consolidated mathematics, supplemented by historical inventories for paths that were absorbed or archived.

The audit used three layers.

1. The current public GitHub tree, the current semi-formalized README, and the raw consolidated *Exponents and q -Series Frontiers* source were inspected at theorem and open-problem level.
2. A preserved recursive ledger of every `*.tex` path under `Analysis/FabiusFunction/docs` at commit `24da985eba70e46d862acdadd476793a428213e` was used to retain coverage of the frozen archive and earlier source-paper layout. The 57-path ledger is included in the companion archive as `corpus_inventory_2026-08-27.txt`.
3. Prior frontier reports in the user’s library were phrase- and formula-searched for the proposed concepts. In particular, Stein–Koopman calculus, transfer determinants, fractional transforms, dyadic-comb interpolation, inverse-Lambert expansions, Appell deconvolution, and spectral zero-divisor theory were treated as occupied territory and are not presented here as new.

The current consolidated q -series volume formulates a *generic natural-boundary conjecture* for its geometric characteristic function and separately asks for *root-of-unity renormalization*. The present report targets precisely that remaining scalar analytic gap. It does not claim to solve the full two-nome finite-law, matrix Mahler, or mixed Prouhet problem.

Status note 1.1 (Meaning of “new”). Throughout this report, *new* means apparently new relative to the audited repository corpus. It is not a claim of worldwide priority. Classical ingredients such as Euler’s sine product, Bernoulli cumulants, q -Pochhammer notation, and radial root-of-unity methods are cited and clearly separated from the deductions made here.

1.2 Result ledger

The principal contributions are as follows.

- R1.** An exact logarithmic and q -Pochhammer representation of the complex geometric sinc product, organized so that cyclotomic resonance is visible coefficient by coefficient. The underlying product and the global spectral q -Pochhammer factorization now have exact Lean counterparts; the local master logarithm remains a manuscript theorem.
- R2.** A two-term radial expansion at every nontrivial root of unity, with an explicit action $\mathcal{A}_\omega$ and constant term $\mathcal{B}_\omega$.
- R3.** Three equivalent descriptions of $\mathcal{A}_\omega$: a resonant zeta series, an infinite spectral dilogarithm, and a cyclic projection integral.
- R4.** A sign theorem for the action and, as a consequence, a proof that the unit circle is a natural boundary for every fixed real $0 < |z| < \pi/2$.
- R5.** A root-of-unity blow-up limit to $\exp(-A_\omega w^{2d})$, including subcritical corrections, the complete order- ε^d layer, and an all-orders Bell-polynomial scheme.

- R6.** A universal analytic-germ theorem that recovers both the sinc result and a root-scaled q -Pochhammer limit.
- R7.** Spectral condensation: among all reciprocal zero power sums, exactly the $2d$ th survives after cyclotomic scaling.
- R8.** Bernoulli–Bell condensation: among all scaled cumulants, exactly the $2d$ th survives, forcing an explicit sparse moment sequence.
- R9.** A limiting Gould–Hopper/Appell family obtained from the inverse moment generating function.
- R10.** Polyharmonic heat-kernel and Stokes-sector interpretations, distinguishing the genuinely dissipative roots $m \equiv 2 \pmod{4}$ from backward/rotated sectors.
- R11.** Formal inverse series in the scaled frequency and in q , with explicit first corrections; sectorial analytic realization and monodromy remain open.
- R12.** Reproducible high-precision tests whose fitted convergence exponents agree with the displayed asymptotic orders.

Status note 1.2 (Current formal and analytic boundary). The repository already defines the complex geometric sinc product and its reciprocal-Gamma reflection factorization. The global spectral q -Pochhammer identity in [equation \(10\)](#) is now machine-checked in Lean; exact names and normalizations are recorded below. No later root-of-unity asymptotic, natural-boundary, condensation, Gould–Hopper, or inverse-branch claim in this report is thereby Lean-verified.

There is also an analytic distinction inside the manuscript. The fixed-index cyclotomic expansions do not by themselves justify summation over an unbounded index. The proof of [theorem 4.3](#) closes this gap with a uniform two-range estimate, splitting at $k \asymp \delta^{-1/2}$ and using geometric decay beyond the cutoff. By contrast, the inverse formulas below record formal coefficients and sectorial targets, not established convergent analytic germs at the boundary.

2 The Fabius–Rvachev interface and its complex q -extension

2.1 The dyadic probability law

Let $V_0, V_1, \dots$ be independent random variables uniformly distributed on $[-1, 1]$. The centered dyadic random series

$$Y_{1/2} = \sum_{j \geq 0} 2^{-(j+1)} V_j \quad (1)$$

has support $[-1, 1]$ and density Rvachev’s up-function up . Its characteristic function, with the convention $\text{sinc}_{\text{rad}} z = \sin z/z$, is

$$\widehat{\text{up}}_{\text{ang}}(-z) = \prod_{j \geq 0} \text{sinc}_{\text{rad}}\left(\frac{z}{2^{j+1}}\right). \quad (2)$$

The Fabius distribution on $[0, 1]$ is obtained by the affine change $X = (Y_{1/2} + 1)/2$. The repository develops this bridge, exact dyadic arithmetic, Thue–Morse derivative signs, moment and cumulant formulas, Legendre expansions, inverse asymptotics, and many related representations. We use those structures only as motivation and normalization.

2.2 The geometric family

For real $0 < q < 1$, define

$$Y_q = (1 - q) \sum_{j \geq 0} q^j V_j. \quad (3)$$

The coefficient sum is one, so Y_q is again supported on $[-1, 1]$. Independence gives

$$\mathbb{E} e^{izY_q} = \prod_{j \geq 0} \text{sinc}_{\text{rad}}((1 - q)q^j z). \quad (4)$$

For complex q there is generally no positive probability law, but the product itself remains a natural analytic object.

Definition 2.1 (Complex q -Fabius–Rvachev sinc product). For $q \in \mathbb{D} = \{q \in \mathbb{C} : |q| < 1\}$ and $z \in \mathbb{C}$, set

$$\Phi(q, z) = \prod_{j=0}^{\infty} \text{sinc}_{\text{rad}}((1 - q)q^j z). \quad (5)$$

Proposition 2.2 (Holomorphy and normalization). *The product in [equation \(5\)](#) converges locally uniformly on $\mathbb{D} \times \mathbb{C}$ and defines a holomorphic function there. Moreover,*

$$\Phi(q, 0) = 1, \quad \Phi(0, z) = \text{sinc}_{\text{rad}} z, \quad \Phi\left(\frac{1}{2}, z\right) = \widehat{\text{up}}_{\text{ang}}(-z).$$

Proof. On a compact set $|q| \leq r < 1$, $|z| \leq R$, the j th argument is $O(r^j)$ uniformly. Since $\text{sinc}_{\text{rad}} w = 1 - w^2/6 + O(w^4)$, the deviation of the j th factor from one is $O(r^{2j})$. The Weierstrass criterion gives locally uniform convergence of the product and of its holomorphic finite approximants. The three normalizations follow directly. $\square$

Remark 2.3 (Exact Lean normalization dictionary). Lean’s existing product uses the frequency coordinate

$$\text{geometricSincProduct}(q, u) = \prod_{j \geq 0} \text{sinc}_{\text{rad}}(\pi q^j u).$$

Consequently the present convention is exactly

$$\Phi(q, z) = \text{geometricSincProduct}\left(q, \frac{(1 - q)z}{\pi}\right).$$

The definition and its module are

```
Fabius.geometricSincProduct in FabiusFunction.GeometricReciprocalGamma.
```

At the dyadic point, the theorem

```
Fabius.centeredSincProduct_eq_rvachevFourierProduct
```

records the further coordinate conversion

$$\Phi\left(\frac{1}{2}, z\right) = \text{centeredSincProduct}(z) = \text{rvachevFourierProduct}\left(\frac{z}{2\pi}\right).$$

Thus the manuscript’s Fourier variable and Lean’s canonical Rvachev product variable differ by the displayed factor 2π .

3 Master logarithmic and q -Pochhammer representations

3.1 The exact logarithmic series

Euler's sine product and its logarithm give

$$\operatorname{sinc}_{\text{rad}} w = \prod_{n \geq 1} \left(1 - \frac{w^2}{\pi^2 n^2}\right), \quad \log \operatorname{sinc}_{\text{rad}} w = - \sum_{k \geq 1} c_k w^{2k}, \quad (6)$$

where

$$c_k = \frac{\zeta(2k)}{k\pi^{2k}} > 0. \quad (7)$$

The logarithm is the branch normalized by $\log \operatorname{sinc}_{\text{rad}} 0 = 0$ in $|w| < \pi$.

Theorem 3.1 (Master logarithm). *On the domain*

$$\Omega = \{(q, z) \in \mathbb{D} \times \mathbb{C} : |(1 - q)z| < \pi\},$$

one has the absolutely and locally uniformly convergent expansion

$$\log \Phi(q, z) = - \sum_{k \geq 1} c_k \frac{(1 - q)^{2k}}{1 - q^{2k}} z^{2k}. \quad (8)$$

Proof. For $(q, z) \in \Omega$, every argument $(1 - q)q^j z$ lies in the zero-free disk $|w| < \pi$. Insert [equation \(6\)](#), sum first over j , and use $\sum_{j \geq 0} q^{2kj} = (1 - q^{2k})^{-1}$. Absolute local uniformity follows from a geometric majorant because $c_k \leq C\pi^{-2k}/k$ and $|(1 - q)z|$ stays below π on compact subsets. $\square$

The coefficient

$$R_k(q) := \frac{(1 - q)^{2k}}{1 - q^{2k}} \quad (9)$$

is simultaneously a geometric power sum, a cumulant multiplier, and the source of every root-of-unity singularity. The entire cyclotomic theory below is a careful organization of the resonances $q^{2k} = 1$.

3.2 A spectral q -Pochhammer factorization

Let $(a; Q)_\infty = \prod_{j \geq 0} (1 - aQ^j)$. Reordering Euler's sine factors gives a second exact representation.

Theorem 3.2 (Infinite spectral q -Pochhammer product). *For every $q \in \mathbb{D}$ and $z \in \mathbb{C}$,*

$$\Phi(q, z) = \prod_{n \geq 1} \left(\frac{(1 - q)^2 z^2}{\pi^2 n^2}; q^2 \right)_\infty. \quad (10)$$

The double product is locally normally convergent.

Proof. Apply Euler's product to each sinc factor:

$$\prod_{j \geq 0} \prod_{n \geq 1} \left(1 - \frac{(1 - q)^2 q^{2j} z^2}{\pi^2 n^2} \right).$$

The sum of absolute first-order deviations is bounded on compact sets by a constant multiple of $\sum_j |q|^{2j} \sum_n n^{-2} < \infty$, so the products may be interchanged. The inner product over j is the stated q^2 -Pochhammer symbol. $\square$

Remark 3.3 (Lean crosswalk: global factorization). The compiled theorem

`Fabius.geometricSincProduct_eq_tprod_complexQPochhammerInf`

in `FabiusFunction.RvachevPochhammerFactorization` states, for every $q \in \mathbb{C}$ with $|q| < 1$ and every $u \in \mathbb{C}$,

$$\text{geometricSincProduct}(q, u) = \prod_{k \geq 0} \left(\frac{u^2}{(k+1)^2}; q^2 \right)_{\infty}.$$

Substituting $u = (1 - q)z/\pi$ and reindexing $n = k + 1$ gives [equation \(10\)](#) exactly. Its proof is global in u , includes $q = 0$ without division by a geometric scale, and remains valid when individual Euler factors vanish. The older dyadic theorem `Fabius.rvachevFourierProduct_eq_tprod_complexQPochhammerInf` is now a direct specialization of this variable- q result.

Remark 3.4 (Why dilogarithms appear). Radial root-of-unity asymptotics of a single q -Pochhammer symbol are governed by a dilogarithmic action and a finite cyclic factor. Formula [equation \(10\)](#) superposes one such action for each spectral point πn . The result is the infinite spectral dilogarithm obtained below. Our direct proof from [equation \(8\)](#) avoids branch bookkeeping and reduces the needed uniformity to the explicit two-range estimate in [theorem 4.3](#).

4 Radial resonance at roots of unity

4.1 Cyclotomic notation

Fix a primitive m th root

$$\omega = \exp\left(\frac{2\pi i h}{m}\right), \quad (h, m) = 1, \quad m \geq 2, \quad (11)$$

and write

$$d = \text{ord}_{\text{mult}}(\omega^2) = \frac{m}{\gcd(m, 2)}, \quad a_{\omega} = 1 - \omega. \quad (12)$$

Thus $\omega^{2k} = 1$ exactly when $d \mid k$. We approach the root radially from inside the disk by

$$q = \omega(1 - \delta), \quad \delta \downarrow 0. \quad (13)$$

Lemma 4.1 (Resonant and nonresonant multipliers). *For fixed $k \geq 1$,*

$$\frac{(1 - q)^{2k}}{1 - q^{2k}} = \frac{a_{\omega}^{2k}}{1 - \omega^{2k}} + O(\delta), \quad d \nmid k, \quad (14)$$

$$\frac{(1 - q)^{2k}}{1 - q^{2k}} = a_{\omega}^{2k} \left[\frac{1}{2k\delta} + \frac{\omega}{1 - \omega} + \frac{2k - 1}{4k} + O(\delta) \right], \quad d \mid k. \quad (15)$$

Both remainders are asserted here only for each fixed k ; no uniformity in an unbounded residue class is part of this lemma.

Proof. The nonresonant denominator tends to the nonzero number $1 - \omega^{2k}$. In the resonant case,

$$\begin{aligned} (1 - q)^{2k} &= a_{\omega}^{2k} \left(1 + 2k \frac{\omega}{a_{\omega}} \delta + O(k^2 \delta^2) \right), \\ 1 - q^{2k} &= 1 - (1 - \delta)^{2k} = 2k\delta - k(2k - 1)\delta^2 + O(k^3 \delta^3). \end{aligned}$$

Divide the two expansions. This proves the two fixed- k statements. Passing from them to a summed expansion requires the separate uniform remainder estimate isolated in the appendix. $\square$

4.2 The radial action and constant term

Definition 4.2 (Cyclotomic action). For $|a_\omega z| < \pi$, define

$$\mathcal{A}_\omega(z) = \sum_{\ell \geq 1} \frac{c_{\ell d}}{2^{\ell d}} a_\omega^{2\ell d} z^{2\ell d}. \quad (16)$$

Also define

$$\mathcal{B}_\omega(z) = - \sum_{\substack{k \geq 1 \\ d \nmid k}} c_k \frac{a_\omega^{2k}}{1 - \omega^{2k}} z^{2k} \quad (17)$$

$$- \sum_{\substack{k \geq 1 \\ d \mid k}} c_k a_\omega^{2k} \left(\frac{\omega}{1 - \omega} + \frac{2k - 1}{4k} \right) z^{2k}. \quad (18)$$

Theorem 4.3 (Uniform two-term radial root-of-unity expansion). *Let K be a compact subset of $\{z : |a_\omega z| < \pi\}$. Uniformly for $z \in K$,*

$$\log \Phi(\omega(1 - \delta), z) = -\frac{\mathcal{A}_\omega(z)}{\delta} + \mathcal{B}_\omega(z) + O_K(\delta). \quad (19)$$

In particular, every nontrivial root of unity carries a full exponential-scale singularity unless $\mathcal{A}_\omega(z) = 0$.

Proof. Put $a = a_\omega$, choose $0 < \rho < \rho_1 < 1$ with $|az| \leq \rho\pi$ on K , and shrink $\delta_0 > 0$ so that $|(a + \omega\delta)z| \leq \rho_1\pi$ for $z \in K$ and $0 \leq \delta \leq \delta_0$. Write the k th multiplier as

$$S_k(\delta) = \frac{(a + \omega\delta)^{2k}}{1 - \omega^{2k}(1 - \delta)^{2k}}.$$

If $d \nmid k$, the finitely many possible phases $\omega^{2k} \neq 1$ give $\inf_{0 \leq t \leq 1} |1 - \omega^{2k}t| > 0$ uniformly in k . Differentiating S_k on the real interval $[0, \delta_0]$, and using the two bounds just chosen, gives

$$c_k |z|^{2k} |S_k(\delta) - S_k(0)| \leq C_K \delta \rho_1^{2k}. \quad (20)$$

Indeed, the derivative contributes at most one factor of k , which cancels the $1/k$ in $c_k \leq \zeta(2)/(k\pi^{2k})$. Summing [equation \(20\)](#) proves the required $O_K(\delta)$ estimate for all nonresonant indices.

Now suppose $d \mid k$, put $n = 2k$ and $b = \omega/a$, and choose the cutoff $N = \lfloor \delta^{-1/2} \rfloor$. Uniformly when $k \leq N$,

$$\begin{aligned} (1 + b\delta)^n &= 1 + nb\delta + O(n^2\delta^2), \\ \frac{1}{1 - (1 - \delta)^n} &= \frac{1}{n\delta} + \frac{n - 1}{2n} + O(n\delta). \end{aligned}$$

Here $n\delta \leq 2\sqrt{\delta}$, so the constants are independent of k and δ . Multiplication yields

$$\frac{S_k(\delta)}{a^{2k}} = \frac{1}{2k\delta} + \frac{\omega}{a} + \frac{2k - 1}{4k} + O(k\delta). \quad (21)$$

After multiplication by $c_k |az|^{2k}$, the summed remainder in [equation \(21\)](#) is bounded by $C_K \delta \sum_{k \leq N} \rho^{2k} = O_K(\delta)$.

It remains only to justify discarding $k > N$. The elementary inequality

$$1 - (1 - \delta)^n \geq 1 - e^{-n\delta} \geq (1 - e^{-1}) \min(1, n\delta)$$

and the ρ_1 bound imply that the actual resonant tail, as well as the three subtracted terms in [equation \(21\)](#), is at most

$$C_K \sum_{k > N} \rho_1^{2k} \left(1 + \frac{1}{k\delta} \right) = O_K(\delta^{-1} \rho_1^{2N}) = o_K(\delta).$$

Thus the fixed-index coefficients may be summed uniformly. Their pole and constant parts are precisely $\mathcal{A}_\omega$ and $\mathcal{B}_\omega$, respectively, which proves [equation \(19\)](#). $\square$

4.3 Dilogarithmic and cyclic descriptions

The action has two additional forms that expose its spectral and finite-cyclic content.

Theorem 4.4 (Spectral dilogarithm). *For $|a_\omega z| < \pi$,*

$$\mathcal{A}_\omega(z) = \frac{1}{2d^2} \sum_{n \geq 1} \text{Li}_2 \left(\left(\frac{a_\omega z}{\pi n} \right)^{2d} \right). \quad (22)$$

The series converges absolutely and locally uniformly.

Proof. Expand $\zeta(2\ell d) = \sum_{n \geq 1} n^{-2\ell d}$ in [equation \(16\)](#), then interchange the absolutely convergent sums:

$$\begin{aligned} \mathcal{A}_\omega(z) &= \sum_{n \geq 1} \sum_{\ell \geq 1} \frac{1}{2(\ell d)^2} \left(\frac{a_\omega z}{\pi n} \right)^{2\ell d} \\ &= \frac{1}{2d^2} \sum_{n \geq 1} \text{Li}_2 \left(\left(\frac{a_\omega z}{\pi n} \right)^{2d} \right). \end{aligned}$$

□

Theorem 4.5 (Cyclic projection formula). *For $|a_\omega z| < \pi$,*

$$\mathcal{A}_\omega(z) = - \int_0^z \left[\frac{1}{d} \sum_{r=0}^{d-1} \log \text{sinc}_{\text{rad}}(a_\omega \omega^r t) \right] \frac{dt}{t}. \quad (23)$$

Here every logarithm is the zero-normalized holomorphic branch from [equation \(6\)](#), and the integral is taken along the straight segment from 0 to z (equivalently, along any path in the disk). The integrand has a holomorphic removable extension at zero.

Proof. Average the logarithmic series:

$$\frac{1}{d} \sum_{r=0}^{d-1} \log \text{sinc}_{\text{rad}}(a_\omega \omega^r t) = - \sum_{k \geq 1} c_k a_\omega^{2k} t^{2k} \left(\frac{1}{d} \sum_{r=0}^{d-1} \omega^{2kr} \right).$$

The parenthesized average is one if $d \mid k$ and zero otherwise. Integrating termwise gives [equation \(16\)](#). □

Remark 4.6. Formula [equation \(23\)](#) is a root-of-unity filter in the scale index. The sinc function is even, so the relevant cycle is generated by ω^2 , not ω ; this explains the compression from order m to $d = m / \gcd(m, 2)$.

5 A natural-boundary theorem

5.1 The sign of the action

The crucial point is that the action cannot vanish on a uniform real-frequency interval.

Lemma 5.1 (Reality and sign of a_ω^{2d}). *The number a_ω^{2d} is real and nonzero. For a primitive m th root,*

$$\text{sgn}(a_\omega^{2d}) = \begin{cases} -1, & m \text{ odd}, \\ (-1)^{d+1}, & m \text{ even}. \end{cases} \quad (24)$$

Thus it is positive exactly when $m \equiv 2 \pmod{4}$.

Proof. Write $\omega = e^{i\theta}$. Since $1 - e^{i\theta} = -2ie^{i\theta/2} \sin(\theta/2)$,

$$a_\omega^{2d} = (-4)^d e^{id\theta} \sin^{2d}(\theta/2).$$

If m is odd then $d = m$ and $e^{id\theta} = 1$, while d is odd. If m is even then $d = m/2$, the primitive numerator h is odd, and $e^{id\theta} = e^{i\pi h} = -1$. The signs follow. $\square$

Theorem 5.2 (Nonvanishing action on a universal interval). *For every primitive root $\omega \neq 1$ and every real z with*

$$0 < |z| < \frac{\pi}{2}, \quad (25)$$

$\mathcal{A}_\omega(z)$ is real, nonzero, and

$$\operatorname{sgn} \mathcal{A}_\omega(z) = \operatorname{sgn}(a_\omega^{2d}). \quad (26)$$

Proof. Because $|a_\omega| \leq 2$, condition [equation \(25\)](#) implies $|a_\omega z / (\pi n)| < 1$ for every $n \geq 1$. By [equation \(24\)](#),

$$\left(\frac{a_\omega z}{\pi n} \right)^{2d} = \sigma x_n, \quad \sigma \in \{-1, 1\}, \quad 0 < x_n < 1.$$

For $0 < x < 1$, $\operatorname{Li}_2(x) > 0$ and $\operatorname{Li}_2(-x) < 0$. Every summand in [equation \(22\)](#) therefore has sign σ , proving the claim. $\square$

5.2 Natural boundary

Theorem 5.3 (Natural boundary on a universal frequency interval). *Fix real z with $0 < |z| < \pi/2$. The holomorphic function*

$$q \mapsto \Phi(q, z), \quad |q| < 1, \quad (27)$$

has the unit circle as a natural boundary. More strongly, it admits no single-valued meromorphic continuation through any nontrivial boundary arc.

Proof. Let $\omega \neq 1$ be any root of unity. By the preceding theorem, $A := \mathcal{A}_\omega(z) \in \mathbb{R} \setminus \{0\}$. From [equation \(19\)](#),

$$\log |\Phi(\omega(1 - \delta), z)| = -\frac{A}{\delta} + O(1). \quad (28)$$

If $A > 0$, the modulus tends to zero faster than every power of δ . A holomorphic continuation through ω would either be nonzero there or have a zero of finite order, both incompatible with this decay. A zero of infinite order would force the continuation to vanish identically near ω , hence everywhere in the connected disk by identity, contradicting $\Phi(0, z) = \operatorname{sinc}_{\text{rad}} z \neq 0$. A meromorphic continuation cannot have a pole because its modulus would then grow at a finite polynomial rate; without a pole it is holomorphic, which is the case just excluded.

If $A < 0$, the modulus grows faster than every power of δ . This is incompatible not only with a holomorphic continuation but also with a meromorphic continuation, whose singularity could have only finite pole order.

Thus every nontrivial root of unity is a nonmeromorphic boundary singularity. Roots of unity are dense on $|q| = 1$, so no open boundary arc can be crossed. $\square$

Remark 5.4 (Scope of the theorem). The theorem gives an explicit, root-independent interval $0 < |z| < \pi/2$. It does not classify exceptional complex frequencies outside that interval. There the action can cross dilogarithm branch cuts, and cancellation between spectral terms is a genuine possibility.

6 Cyclotomic blow-ups

The fixed-frequency expansion diverges like δ^{-1} . A nontrivial local limit appears when the frequency shrinks at the exact resonant rate.

6.1 Leading blow-up theorem

Set

$$q_\varepsilon = \omega(1 - \varepsilon^d), \quad z = \sqrt{\varepsilon} w, \quad \varepsilon \downarrow 0, \quad (29)$$

and define

$$A_\omega = \frac{c_d a_\omega^{2d}}{2d} = \frac{\zeta(2d) a_\omega^{2d}}{2d^2 \pi^{2d}}. \quad (30)$$

Lemma 6.1 (Uniform scaled-tail bound). *For every compact set $W \subset \mathbb{C}$ there are $C_W > 0$ and $\varepsilon_0 > 0$ such that, for every integer $M \geq d$, every $w \in W$, and $0 < \varepsilon \leq \varepsilon_0$, the tail of the master logarithm satisfies*

$$\sum_{k \geq M} c_k \varepsilon^k |w|^{2k} \left| \frac{(a_\omega + \omega \varepsilon^d)^{2k}}{1 - \omega^{2k} (1 - \varepsilon^d)^{2k}} \right| \leq C_W \varepsilon^{M-d}. \quad (31)$$

Proof. Put $s = \varepsilon^d$. If $d \mid k$, then $1 - (1 - s)^{2k} \geq s$. If $d \nmid k$, the finitely many phases $\omega^{2k} \neq 1$ keep $|1 - \omega^{2k} (1 - s)^{2k}|$ uniformly away from zero. After reducing ε_0 if necessary, both cases are therefore bounded below by $c_\omega s$ for a fixed $c_\omega > 0$. If $|w| \leq R$ on W , then $c_k \leq \zeta(2)/(k\pi^{2k})$ turns the left side into at most

$$C \varepsilon^{-d} \sum_{k \geq M} \frac{1}{k} \left(\frac{|a_\omega + \omega \varepsilon^d|^2 R^2}{\pi^2} \varepsilon \right)^k.$$

The parenthesized base is at most $B_W \varepsilon < 1/2$ for small ε . Summing the geometric tail proves [equation \(31\)](#). $\square$

Theorem 6.2 (Cyclotomic blow-up limit). *Locally uniformly for $w \in \mathbb{C}$,*

$$\boxed{\Phi(q_\varepsilon, \sqrt{\varepsilon} w) \longrightarrow \exp(-A_\omega w^{2d})}. \quad (32)$$

Proof. In [equation \(8\)](#), the k th term acquires the factor ε^k . If $d \nmid k$, the multiplier [equation \(9\)](#) remains bounded and the term tends to zero. If $k = \ell d$, its multiplier is asymptotic to $a_\omega^{2k}/(2k\varepsilon^d)$, so the term has order $\varepsilon^{(\ell-1)d}$. Only $k = d$ survives, giving $-A_\omega w^{2d}$. Taking $M = d + 1$ in [equation \(31\)](#) makes every omitted term $O_W(\varepsilon)$, which proves local uniformity and permits exponentiation. $\square$

6.2 Subcritical and order- d corrections

For $1 \leq k < d$ define

$$C_{k,\omega} = c_k \frac{a_\omega^{2k}}{1 - \omega^{2k}}, \quad (33)$$

and set

$$\eta_\omega = \frac{2d\omega}{1 - \omega} + \frac{2d - 1}{2}, \quad D_\omega = \frac{c_{2d} a_\omega^{4d}}{4d}. \quad (34)$$

Theorem 6.3 (First complete correction layers). *Uniformly on compact w -sets,*

$$\log \Phi(q_\varepsilon, \sqrt{\varepsilon} w) = -A_\omega w^{2d} - \sum_{k=1}^{d-1} C_{k,\omega} \varepsilon^k w^{2k} \quad (35)$$

$$- \varepsilon^d \left(A_\omega \eta_\omega w^{2d} + D_\omega w^{4d} \right) + O(\varepsilon^{d+1}). \quad (36)$$

Proof. For $k < d$ there is no resonance, so the constant term of the multiplier gives [equation \(33\)](#). At total order ε^d , two and only two resonant sources contribute. The next term in the $k = d$ multiplier is the constant in [equation \(15\)](#), producing $A_\omega \eta_\omega w^{2d}$. The leading pole of the $k = 2d$ multiplier produces $D_\omega w^{4d}$. Every omitted nonresonant term begins at $k = d + 1$, and every later resonant term begins at order at least $2d$. Taylor's theorem handles the finitely many displayed indices, while [equation \(31\)](#) with $M = 2d + 1$ handles the infinite remainder, giving $O(\varepsilon^{d+1})$. $\square$

6.3 All orders and Bell polynomials

Define the Laurent germ

$$\mathcal{R}_k(\varepsilon) = \frac{(a_\omega + \omega \varepsilon^d)^{2k}}{1 - \omega^{2k}(1 - \varepsilon^d)^{2k}}. \quad (37)$$

For $d \nmid k$ it is analytic at zero; for $d \mid k$ it has a simple pole in ε^d .

Theorem 6.4 (All-orders logarithmic expansion). *There are even polynomials $L_n(w)$ such that, as a locally uniform asymptotic expansion,*

$$\log \Phi(q_\varepsilon, \sqrt{\varepsilon} w) \sim -A_\omega w^{2d} + \sum_{n \geq 1} L_n(w) \varepsilon^n, \quad (38)$$

where the coefficient is the finite sum

$$L_n(w) = - \sum_{k=1}^{n+d} c_k w^{2k} \left[\varepsilon^{n-k} \right] \mathcal{R}_k(\varepsilon). \quad (39)$$

Consequently,

$$\Phi(q_\varepsilon, \sqrt{\varepsilon} w) \sim e^{-A_\omega w^{2d}} \left(1 + \sum_{n \geq 1} P_n(w) \varepsilon^n \right), \quad (40)$$

with

$$P_n(w) = \frac{1}{n!} B_n(1!L_1(w), 2!L_2(w), \dots, n!L_n(w)), \quad (41)$$

where B_n is the complete exponential Bell polynomial.

Proof. Substitute equation (37) into the master logarithm. The coefficient of ε^n in the k th term is the displayed Laurent coefficient with index $n - k$. Since the lowest possible exponent in $\mathcal{R}_k$ is $-d$, only $k \leq n + d$ can contribute, so the sum is finite. For a truncation through order n , Taylor's theorem applies to those finitely many Laurent germs, and equation (31) with $M = n + d + 1$ bounds the remaining infinite sum by $O_W(\varepsilon^{n+1})$. This proves the compact-uniform Poincaré expansion. Formula equation (41) is the standard exponential formula for exponentiating a formal series. $\square$

Remark 6.5. This is a second Bell structure in the theory. The repository's Bell polynomials encode moments from cumulants; here the same combinatorial transform encodes multiplicative cyclotomic corrections from logarithmic ones. The two uses meet in section 9.

6.4 Four small root orders

Example 6.6 (The root $\omega = -1$). Here $m = 2$, $d = 1$, and $A_{-1} = 1/3$. Thus

$$\log \Phi(-1 + \varepsilon, \sqrt{\varepsilon} w) = -\frac{w^2}{3} + \varepsilon \left(\frac{w^2}{6} - \frac{w^4}{45} \right) + O(\varepsilon^2). \quad (42)$$

The first cyclotomic limit is Gaussian.

Example 6.7 (Primitive cube root). For $\omega = e^{2\pi i/3}$, $d = 3$ and $A_\omega = -1/630$. Moreover

$$C_{1,\omega} = -\frac{i}{2\sqrt{3}}, \quad C_{2,\omega} = -\frac{i}{20\sqrt{3}}.$$

Hence

$$\log \Phi(\omega(1 - \varepsilon^3), \sqrt{\varepsilon} w) = \frac{w^6}{630} + \frac{i\varepsilon w^2}{2\sqrt{3}} + \frac{i\varepsilon^2 w^4}{20\sqrt{3}} + O(\varepsilon^3). \quad (43)$$

The real-axis limit grows, so its transform interpretation belongs to rotated Stokes sectors.

Example 6.8 (Fourth root). For $\omega = i$, $d = 2$, $A_\omega = -1/180$, and $C_{1,\omega} = -i/6$. Therefore

$$\log \Phi(i(1 - \varepsilon^2), \sqrt{\varepsilon} w) = \frac{w^4}{180} + \frac{i\varepsilon w^2}{6} \quad (44)$$

$$- \varepsilon^2 \left[\left(\frac{1}{360} - \frac{i}{90} \right) w^4 + \frac{w^8}{18900} \right] + O(\varepsilon^3). \quad (45)$$

Example 6.9 (Primitive sixth root). For $\omega = e^{\pi i/3}$, $d = 3$, $A_\omega = 1/17010$,

$$C_{1,\omega} = -\frac{i}{6\sqrt{3}}, \quad C_{2,\omega} = \frac{i\sqrt{3}}{540}.$$

Thus

$$\log \Phi(\omega(1 - \varepsilon^3), \sqrt{\varepsilon} w) = -\frac{w^6}{17010} + \frac{i\varepsilon w^2}{6\sqrt{3}} - \frac{i\sqrt{3}\varepsilon^2 w^4}{540} + O(\varepsilon^3). \quad (46)$$

This is the first genuinely higher-order dissipative limit.

7 A universal analytic-germ principle

The sinc product is one instance of a more general mechanism.

Theorem 7.1 (Universal resonant germ). *Let h be entire, with $h(0) = 1$, and choose the local logarithm*

$$\log h(z) = \sum_{n \geq 1} b_n z^n. \quad (47)$$

For $|q| < 1$ define

$$\mathcal{P}_h(q, z) = \prod_{j \geq 0} h((1 - q)q^j z). \quad (48)$$

Fix a root $\omega \neq 1$, and let

$$r = \min \{n \geq 1 : b_n \neq 0 \text{ and } \omega^n = 1\}. \quad (49)$$

Assume this set is nonempty. Along the radial approach $q = \omega(1 - \delta)$ with $\delta > 0$ and $\delta \downarrow 0$, interpret $\delta^{1/r}$ as the positive real root. Then, locally uniformly for w ,

$$\boxed{\mathcal{P}_h(\omega(1 - \delta), \delta^{1/r} w) \longrightarrow \exp \left(\frac{b_r (1 - \omega)^r}{r} w^r \right)}. \quad (50)$$

Proof. The logarithm of [equation \(48\)](#) is

$$\sum_{n \geq 1} b_n \frac{(1 - q)^n}{1 - q^n} z^n.$$

Choose a zero-free disk about zero for h . Cauchy's estimate for its analytic logarithm gives $|b_n| \leq C R_0^{-n}$ for some $R_0 > 0$. For nonresonant n , the finitely many possible phases $\omega^n \neq 1$ keep the denominator uniformly away from zero, and the scaled term is $O(\delta^{n/r})$. For resonant n , $1 - (1 - \delta)^n \geq \delta$, while the $n = r$ term converges to

$$b_r w^r \frac{(a_\omega + \omega \delta)^r \delta}{1 - (1 - \delta)^r} \longrightarrow \frac{b_r a_\omega^r}{r} w^r.$$

Every resonant coefficient below r vanishes by its definition. Finally, on a compact w -set the entire tail $n > r$ is bounded, in both resonance classes, by

$$C \delta^{-1} \sum_{n > r} (B \delta^{1/r})^n = O(\delta^{1/r}).$$

Thus exactly the $n = r$ term survives, uniformly on compact sets. Exponentiating gives [equation \(50\)](#). $\square$

Corollary 7.2 (Sinc and q -Pochhammer specializations). *For $h(z) = \text{sinc}_{\text{rad}} z$, only even coefficients occur and the first resonance is $r = 2d$, recovering [equation \(32\)](#). For $h(z) = 1 - z$ and a primitive m th root, $r = m$ and*

$$\prod_{j \geq 0} \left(1 - (1 - q)q^j \delta^{1/m} w\right) \longrightarrow \exp \left(-\frac{(1 - \omega)^m}{m^2} w^m \right). \quad (51)$$

Remark 7.3. The theorem separates three ingredients: the local Taylor support of $\log h$, the cyclic order of ω , and the geometric denominator $1 - q^n$. It therefore applies to many q -analog kernels, not merely to the Fabius family.

8 Spectral zero condensation

8.1 The zero cloud

For $0 < |q| < 1$, every sinc factor contributes the factor-labelled zeros

$$z_{\ell,j}(q) = \frac{\pi \ell}{(1 - q)q^j}, \quad \ell \in \mathbb{Z} \setminus \{0\}, \quad j \geq 0. \quad (52)$$

The division-free form

$$(1 - q)q^j z = \pi \ell$$

is valid also at $q = 0$ and is the correct total statement there. Distinct pairs (ℓ, j) may describe the same geometric zero; throughout this section reciprocal sums count the Euler factors with their resulting multiplicity. For positive ℓ , define the reciprocal even power sum

$$S_k(q) = \sum_{j \geq 0} \sum_{\ell \geq 1} z_{\ell,j}(q)^{-2k}. \quad (53)$$

Proposition 8.1 (Exact reciprocal zero moments). *For every $k \geq 1$,*

$$S_k(q) = \frac{\zeta(2k)}{\pi^{2k}} \frac{(1 - q)^{2k}}{1 - q^{2k}}. \quad (54)$$

Moreover, the master logarithm is the canonical zero-moment expansion

$$\log \Phi(q, z) = - \sum_{k \geq 1} \frac{S_k(q)}{k} z^{2k}. \quad (55)$$

Proof. Insert [equation \(52\)](#), then separate the geometric and zeta sums. The exact identity

$$\frac{S_k(q)}{k} = c_k \frac{(1 - q)^{2k}}{1 - q^{2k}}$$

makes [equation \(55\)](#) identical to the master logarithm. $\square$

Theorem 8.2 (One surviving spectral moment). *Under the cyclotomic scaling [equation \(29\)](#),*

$$\varepsilon^k S_k(q_\varepsilon) \longrightarrow \begin{cases} \frac{\zeta(2d)}{\pi^{2d}} \frac{a_\omega^{2d}}{2d}, & k = d, \\ 0, & k \neq d. \end{cases} \quad (56)$$

Proof. If $d \nmid k$, the multiplier in [equation \(54\)](#) remains bounded. If $k = \ell d$, it grows like ε^{-d} , so multiplication by ε^k gives $\varepsilon^{(\ell-1)d}$. Only $\ell = 1$ survives. $\square$

Remark 8.3 (Escaping zeros leave an exponential trace). The limiting entire function $e^{-A_\omega w^{2d}}$ has no zeros. The zero cloud of the rescaled sinc product escapes to infinity, but its $2d$ th reciprocal moment remains finite and becomes the exponential factor in the Hadamard limit. All other reciprocal moments disappear.

9 Bernoulli cumulants, Bell moments, and Appell collapse

9.1 Complex moment functional

For real $0 < q < 1$, the moment generating function of Y_q is

$$M(q, t) = \mathbb{E}e^{tY_q} = \Phi(q, it) = \prod_{j \geq 0} \frac{\sinh((1-q)q^j t)}{(1-q)q^j t}. \quad (57)$$

For complex q , [equation \(57\)](#) defines an entire generating function and hence a complex moment functional. The classical Bernoulli expansion gives

$$\kappa_{2k}(q) = \frac{2^{2k-1} B_{2k}}{k} \frac{(1-q)^{2k}}{1-q^{2k}}, \quad \kappa_{2k+1}(q) = 0. \quad (58)$$

Theorem 9.1 (Cyclotomic cumulant condensation). *Let $Z_\varepsilon = \sqrt{\varepsilon} Y_{q_\varepsilon}$ at the level of formal moments. Then*

$$\kappa_n(Z_\varepsilon) \longrightarrow 0 \quad (n \neq 2d), \quad (59)$$

and

$$\boxed{\kappa_{2d}(Z_\varepsilon) \longrightarrow K_{2d,\omega} = \frac{2^{2d-2} B_{2d} a_\omega^{2d}}{d^2}.} \quad (60)$$

Equivalently, coefficientwise near $t = 0$,

$$M(q_\varepsilon, \sqrt{\varepsilon} t) \longrightarrow \exp(B_\omega t^{2d}), \quad B_\omega = (-1)^{d+1} A_\omega = \frac{K_{2d,\omega}}{(2d)!}. \quad (61)$$

Proof. Apply the same resonance count as in [equation \(56\)](#) to [equation \(58\)](#), remembering that scaling by $\sqrt{\varepsilon}$ multiplies the n th cumulant by $\varepsilon^{n/2}$. The identity between B_ω and A_ω follows from Euler's formula for B_{2d} in terms of $\zeta(2d)$. $\square$

Corollary 9.2 (Sparse Bell moment limit). *Let $m_n(q)$ be the moments generated by $M(q, t)$. Then*

$$\varepsilon^{n/2} m_n(q_\varepsilon) \longrightarrow 0 \quad \text{unless } n = 2dr, \quad (62)$$

and for $r \geq 0$,

$$\boxed{\varepsilon^{dr} m_{2dr}(q_\varepsilon) \longrightarrow \frac{(2dr)!}{r!} B_\omega^r.} \quad (63)$$

Proof. Expand $\exp(B_\omega t^{2d})$. Equivalently, apply the complete Bell polynomial to a cumulant list with exactly one nonzero entry. $\square$

9.2 Limiting Appell family

Let $\mathcal{A}_n^{(q)}$ denote the inverse-moment Appell polynomials generated by

$$\frac{e^{xt}}{M(q, t)} = \sum_{n \geq 0} \mathcal{A}_n^{(q)}(x) \frac{t^n}{n!}. \quad (64)$$

Define the scaled family

$$\tilde{\mathcal{A}}_{n,\varepsilon}(x) = \varepsilon^{n/2} \mathcal{A}_n^{(q_\varepsilon)}(x/\sqrt{\varepsilon}). \quad (65)$$

Theorem 9.3 (Gould–Hopper cyclotomic limit). *Coefficientwise as $\varepsilon \downarrow 0$,*

$$\tilde{\mathcal{A}}_{n,\varepsilon}(x) \longrightarrow H_n^{(d,B\omega)}(x), \quad (66)$$

where

$$\sum_{n \geq 0} H_n^{(d,B)}(x) \frac{t^n}{n!} = e^{xt - Bt^{2d}} \quad (67)$$

and explicitly

$$H_n^{(d,B)}(x) = n! \sum_{r=0}^{\lfloor n/(2d) \rfloor} \frac{(-B)^r x^{n-2dr}}{r!(n-2dr)!}. \quad (68)$$

These polynomials satisfy

$$H_n^{(d,B)} = e^{-BD^{2d}} x^n, \quad H_{n+1}^{(d,B)} = \left(x - 2dBD^{2d-1}\right) H_n^{(d,B)}, \quad D = \frac{d}{dx}. \quad (69)$$

Proof. The generating function of equation (65) is $e^{xt}/M(q_\varepsilon, \sqrt{\varepsilon}t)$. Apply equation (61). Expanding the limiting exponential gives equation (68); differentiating its generating function gives the operator identities. $\square$

10 Polyharmonic heat symbols and Stokes sectors

10.1 Dissipative roots

By equation (24), $A_\omega > 0$ exactly when the primitive root order satisfies $m \equiv 2 \pmod{4}$. In that case the cyclotomic limit decays on the real frequency axis.

Definition 10.1 (Cyclotomic polyharmonic kernel). For $A > 0$, $d \geq 1$, and $\tau > 0$, set

$$K_{d,A}(x, \tau) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-\tau A \xi^{2d} - ix\xi} d\xi. \quad (70)$$

Proposition 10.2 (Polyharmonic semigroup). *The kernel is Schwartz in x , has total mass one, and solves*

$$\partial_\tau K_{d,A} = (-1)^{d+1} A \partial_x^{2d} K_{d,A}, \quad K_{d,A}(\cdot, \tau) \rightarrow \delta_0 \quad (\tau \downarrow 0) \quad (71)$$

in the distributional sense. It obeys the scaling law

$$K_{d,A}(x, \tau) = (A\tau)^{-1/(2d)} K_{d,1}\left(x(A\tau)^{-1/(2d)}, 1\right). \quad (72)$$

For $d = 1$ it is Gaussian; for $d \geq 2$ it necessarily changes sign.

Proof. Fourier inversion gives smoothness, mass, the PDE, and scaling. Suppose $d \geq 2$ and $e^{-A\xi^{2d}}$ were a characteristic function of a positive probability measure. Its second derivative at zero is zero, which would force variance zero; the measure would be a point mass and its characteristic function identically one, a contradiction. $\square$

Corollary 10.3 (The sixth-order Rvachev cyclotomic heat limit). *At a primitive sixth root, $d = 3$ and $A_\omega = 1/17010$. The rescaled complex q -Fabius transform converges to the Fourier multiplier of*

$$\partial_\tau u = \frac{1}{17010} \partial_x^6 u. \quad (73)$$

The associated fundamental solution has unit mass but alternating sign.

10.2 Backward roots and contour rotation

If $A_\omega < 0$ (all odd m and all $m \equiv 0 \pmod{4}$), the real-axis symbol $e^{-A_\omega w^{2d}}$ grows. It should not be interpreted as a tempered Fourier multiplier on $\mathbb{R}$. It is nevertheless an entire function with canonical decaying sectors

$$\left\{ w : \Re(A_\omega w^{2d}) > 0 \right\}. \quad (74)$$

More precisely, let Γ be a piecewise smooth contour with two unbounded ends and suppose that, outside a compact set, $\Re(A_\omega w^{2d}) \geq c|w|^{2d}$ for some $c > 0$. Then

$$K_\Gamma(x, \tau) = \frac{1}{2\pi} \int_\Gamma e^{-\tau A_\omega w^{2d} - i x w} dw$$

converges absolutely and locally uniformly in $x \in \mathbb{C}$ for $\tau > 0$. The degree- $2d$ decay dominates the linear xw term, so differentiation under the integral is justified and K_Γ solves the same polyharmonic equation as in [equation \(71\)](#). The sector boundaries $\Re(A_\omega w^{2d}) = 0$ are Stokes lines. This produces a clean arithmetic/Stokes dichotomy: the residue class of m modulo four determines whether the canonical radial blow-up is forward dissipative or backward/rotated.

Problem 10.4 (Uniform Stokes connection). *Construct a single Riemann–Hilbert or steepest-descent description that connects the real dissipative kernels to the rotated kernels across the sign change of A_ω , and determine the Stokes multipliers induced by the finite cyclic factor hidden in $\mathcal{B}_\omega$.*

11 Inverse cyclotomic branches

Throughout this section, $O(\varepsilon^N)$ and $O(z^N)$ denote formal order: the omitted series belongs to the corresponding principal ideal in the indicated formal power-series ring. No analytic remainder estimate is implicit.

11.1 Inverse frequency at fixed radial scaling

Fix a nonzero target $y \in \mathbb{C}$ and a logarithm branch $\ell = \text{Log } y$ with $\ell \neq 0$:

$$\ell = \text{Log } y. \quad (75)$$

Choose one of the $2d$ roots $w_0 \neq 0$ satisfying

$$-A_\omega w_0^{2d} = \ell. \quad (76)$$

Proposition 11.1 (Formal inverse frequency series). *There is a unique formal series $w(\varepsilon) \in w_0 + \varepsilon\mathbb{C}[[\varepsilon]]$ which solves, coefficient by coefficient in the formal expansion [equation \(38\)](#),*

$$\Phi(q_\varepsilon, \sqrt{\varepsilon} w(\varepsilon)) = y, \quad w(0) = w_0. \quad (77)$$

If $d \geq 2$, then

$$w(\varepsilon) = w_0 - \frac{C_{1,\omega}}{2dA_\omega} w_0^{3-2d} \varepsilon + O(\varepsilon^2). \quad (78)$$

For $d = 1$,

$$w(\varepsilon) = w_0 - \frac{\eta_\omega}{2} w_0 \varepsilon - \frac{D_\omega}{2A_\omega} w_0^3 \varepsilon + O(\varepsilon^2). \quad (79)$$

All higher coefficients are obtained recursively from [equation \(39\)](#) and partial Bell polynomials.

Proof. Use the prescribed logarithm value ℓ and substitute a formal series for w in [equation \(38\)](#). The derivative of the constant-term equation with respect to w is $-2dA_\omega w_0^{2d-1} \neq 0$, hence it is a unit and determines each successive coefficient triangularly (equivalently, apply the formal implicit-function theorem). At order ε , compare the derivative contribution with the $-C_{1,\omega}w^2$ term when $d \geq 2$. For $d = 1$, use the complete order-one expression in [equation \(36\)](#). $\square$

Remark 11.2 (Analytic realization remains open). The proposition is formal: it does not assert a convergent germ in a full complex neighborhood of $\varepsilon = 0$. The denominators $1 - (1 - \varepsilon^d)^{2k}$ have complex singularities accumulating at the origin as $k \rightarrow \infty$. Thus the direct analytic implicit-function argument is not available without proving cancellations among these singularities. Any analytic realization must supply such control, or else be formulated sectorially or asymptotically with a uniform remainder estimate.

11.2 Inverse q at small frequency

The more unusual inverse asks for q as a function of z while the transform value is held fixed.

Proposition 11.3 (Formal inverse q -series at a root of unity). *Assume $d \geq 2$, fix y and ℓ as above, and choose the root-centered coordinate*

$$q(z) = \omega(1 - \delta(z)). \quad (80)$$

There is a unique formal series

$$\rho(s) \in -\frac{A_\omega}{\ell} + s\mathbb{C}[[s]], \quad \delta(z) = z^{2d}\rho(z^2),$$

such that, after substitution into the master logarithm on the prescribed logarithm sheet, the identity

$$\Phi(q(z), z) = y \quad (81)$$

holds coefficient by coefficient. Its first terms are

$$\delta(z) = -\frac{A_\omega}{\ell}z^{2d} \left(1 - \frac{C_{1,\omega}}{\ell}z^2 + O(z^4) \right), \quad (82)$$

and therefore

$$q(z) = \omega \left[1 + \frac{A_\omega}{\ell}z^{2d} - \frac{A_\omega C_{1,\omega}}{\ell^2}z^{2d+2} + O(z^{2d+4}) \right]. \quad (83)$$

Proof. Write $\delta = \rho z^{2d}$. Coefficientwise substitution in the master series has constant term $-A_\omega/\rho$. The equation $-A_\omega/\rho = \ell$ has the simple solution $\rho_0 = -A_\omega/\ell$, since the derivative A_ω/ρ_0^2 is nonzero. The formal implicit-function theorem, or direct triangular coefficient recursion, gives $\rho(z^2)$. Expanding

$$\log \Phi = -\frac{A_\omega z^{2d}}{\delta} - C_{1,\omega}z^2 + O(z^4) + O(z^{2d})$$

and matching the coefficient of z^2 yields [equation \(82\)](#). $\square$

Remark 11.4 (No convergent boundary germ is claimed). The formal series is meaningful because every requested coefficient involves only finitely many orders of the master expansion. It does not by itself define an analytic germ through $z = 0$: the root-of-unity singular loci of the individual denominators accumulate there. Establishing sectorial sums, their domains, and their Stokes transitions is a separate problem.

Remark 11.5 (Formal branch geometry). For the inverse-frequency problem, each fixed root ω and logarithm value $\ell + 2\pi i k$ has $2d$ choices of w_0 , and each choice produces one formal series. For the inverse- q problem, fixing ω and a nonzero logarithm value determines a unique formal series ρ . Turning either algebraic indexing into analytic continuation or monodromy requires sectorial realizations and a careful separation of logarithm-sheet changes from critical-value branching; neither is asserted here.

12 Numerical experiments

12.1 Evaluation strategy

Direct products are poorly conditioned near roots of unity. The companion script evaluates the absolutely convergent master series [equation \(8\)](#) with 80 decimal digits, and computes actions and constants from their own convergent series. No numerical quadrature or random sampling is used. The four main diagnostics are:

1. the residual in the two-term fixed-frequency radial expansion;
2. the residual after removing every displayed blow-up correction through order ε^d ;
3. the error in the surviving Bernoulli cumulant and Bell moments;
4. the errors in the inverse frequency and inverse q branches.

12.2 Radial expansion

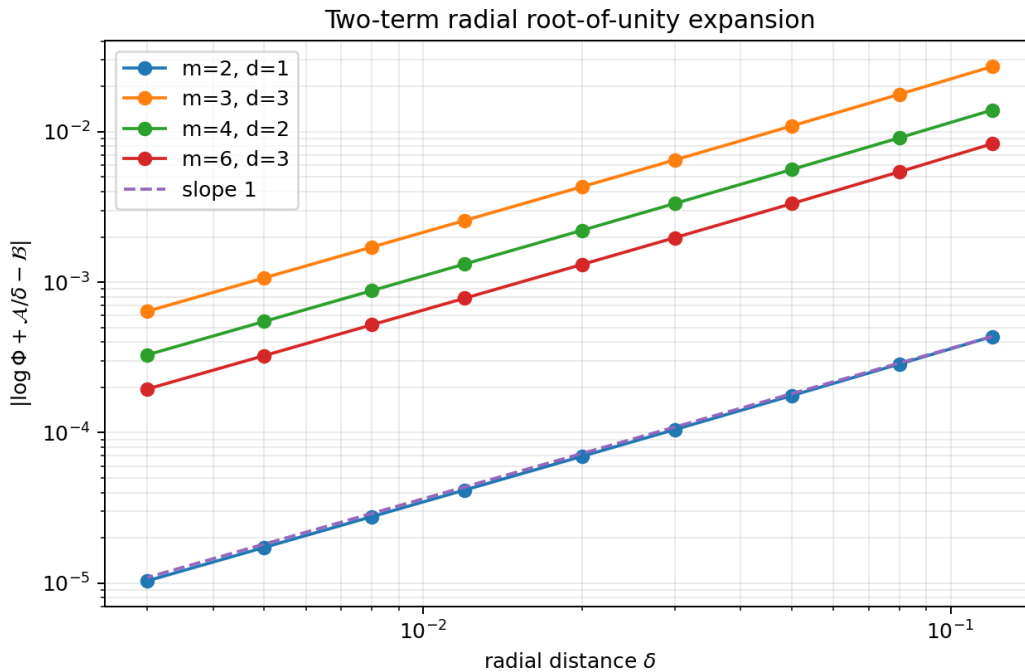


Figure 1: For $z = 0.8$, the residual $|\log \Phi + \mathcal{A}/\delta - \mathcal{B}|$ is asymptotically linear in δ at four root orders. The fitted slopes over the five smallest radial distances are all between 1.0043 and 1.0045.

The calculation tests both the divergent action and the finite constant term. It is notably sensitive to the $(2k - 1)/(4k)$ term in [equation \(15\)](#); omitting that term changes the residual from $O(\delta)$ to $O(1)$.

12.3 Blow-up and inverse errors

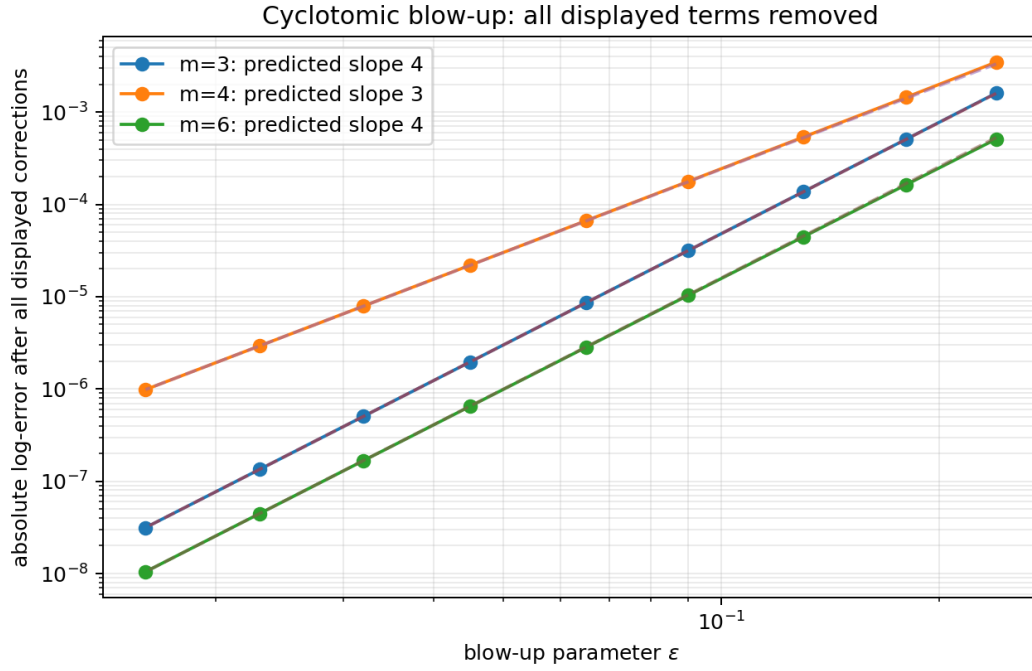


Figure 2: Absolute logarithmic error after removing the limit, all subcritical terms, and the complete order- ε^d layer. The predicted next exponents are $d + 1$: four for $m = 3, 6$ and three for $m = 4$.

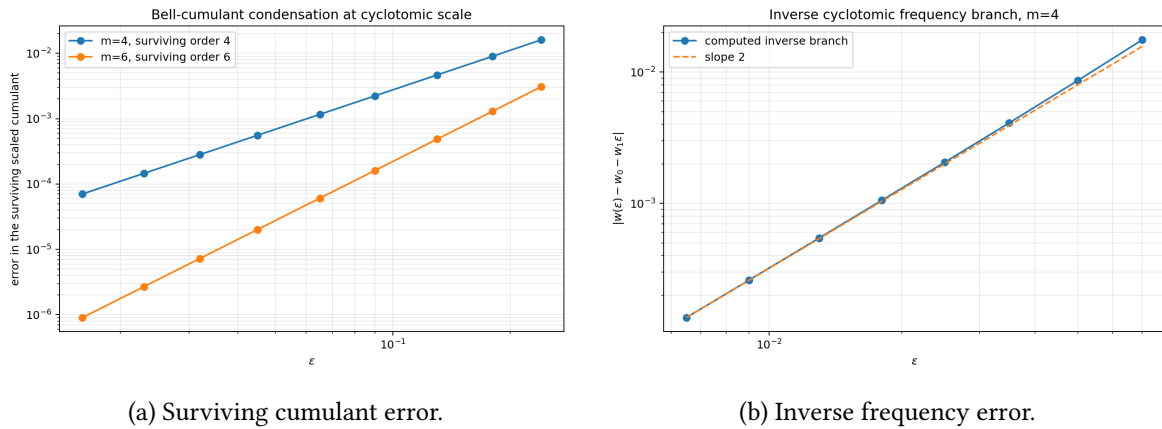


Figure 3: The Bernoulli cumulant converges to the unique order- $2d$ limit; the first-order inverse frequency formula has the predicted quadratic remainder.

12.4 Fitted exponents

Table 1: Log–log slopes fitted to the five smallest parameters. The final column gives the absolute error at the smallest tested parameter.

diagnostic	predicted slope	fitted slope	last error
fixed radial, $m = 2$	1	1.0043	$1.034e - 05$
fixed radial, $m = 3$	1	1.0044	$6.407e - 04$
fixed radial, $m = 4$	1	1.0044	$3.289e - 04$
fixed radial, $m = 6$	1	1.0045	$1.949e - 04$
blow-up, $m = 3$	4	4.0001	$3.146e - 08$
blow-up, $m = 4$	3	3.0041	$9.846e - 07$
blow-up, $m = 6$	4	3.9934	$1.045e - 08$
inverse frequency, $m = 4$	2	2.0210	$1.350e - 04$
inverse q -branch, $m = 4$	8	7.9960	$3.534e - 08$

The inverse q experiment at $\omega = i$ fixes $y = 1.1$ and solves for $q(z) = i(1 - \delta(z))$. The leading approximation has error $O(z^6)$; after the $C_{1,\omega}$ correction, the fitted exponent is 7.9960, matching the predicted $O(z^8)$ remainder. At $z = 0.145$, the computed local q -coordinate was

$$\delta = 2.57314407304007927 \times 10^{-5} + 9.47348632511157983 \times 10^{-7}i,$$

while the two-term absolute error was 3.53×10^{-8} .

Status note 12.1 (Role of computation). The numerical experiments do not substitute for proofs. They check constants, signs, and predicted asymptotic orders, and they are especially useful for detecting missing resonant correction terms. In particular, they do not prove the uniform radial estimate established analytically above, convergence of either formal inverse series, or any Lean theorem. Only the global spectral q -Pochhammer factorization has the exact Lean crosswalk recorded above; the remaining cyclotomic assertions retain manuscript or frontier status according to the boundary note at the start of the report.

13 Connections with Thue–Morse and cyclic digital structure

The original dyadic theory combines two kinds of scale symmetry. The sinc product uses geometric scales 2^{-j} , while the Thue–Morse polynomial $\prod_{j < N} (1 - z^{2^j})$ encodes binary digit parity. The cyclotomic analysis introduces a third viewpoint: a periodic character in the *scale index* j .

At $q = \omega(1 - \delta)$, the leading phases $q^j \approx \omega^j$ repeat with period m . Because sinc is even, the logarithm sees ω^{2j} and the effective period is d . The cyclic projector in [equation \(23\)](#) is precisely the finite Fourier projection onto the resonant scale characters. At $\omega = -1$, the period-two character collapses to $d = 1$ and produces the Gaussian limit. Higher roots produce higher-order heat symbols.

This does not identify the cyclotomic scale character with the Thue–Morse sequence itself: Thue–Morse parity is generated by binary digits of the *summation index*, whereas ω^j is periodic in the *scale position*. A promising bridge is to combine the two characters in a two-nome product, exactly the larger problem left open in the repository.

Problem 13.1 (Thue–Morse/cyclotomic coupling). *For finite products carrying both a Thue–Morse sign and a scale phase ω^j , determine the resonant cumulant order, the analogue of $\mathcal{A}_\omega$, and whether Prouhet cancellation can suppress the leading spectral dilogarithm. Classify the resulting limit symbols by the joint character group.*

14 Further conjectures and research directions

14.1 Maximal natural-boundary domain

Problem 14.1 (Generic natural boundary and action-zero control). *Determine a precise exceptional set of frequencies for which enough root-of-unity actions $\mathcal{A}_\omega(z)$ vanish to obstruct the natural-boundary argument. Prove at least a residual or full-measure generic statement, and identify additional uniform arithmetic hypotheses that would give an explicit complement. A locally finite exceptional set is not assumed: a countable union of discrete action-zero sets need not be locally finite.*

The theorem proved here covers $0 < |z| < \pi/2$ uniformly. Outside that interval, the dilogarithm arguments may leave the unit disk or encounter branch cuts, and the sign argument is unavailable. On its defining disk each action is already a nonzero analytic germ, since $\mathcal{A}_\omega(z) = A_\omega z^{2d} + O(z^{4d})$ with $A_\omega \neq 0$, so its zeros there are discrete. The genuine target is simultaneous control of these zero sets across a dense family of roots, on compatible larger frequency domains; a Baire-category argument might then yield generic natural boundary, while arithmetic control of intersections could sharpen “generic” to an explicit complement.

14.2 Cyclic quantum dilogarithm in the finite term

The leading action is dilogarithmic, while standard root-of-unity asymptotics of q -Pochhammer symbols also contain a finite cyclic quantum dilogarithm. Formula [equations \(17\) and \(18\)](#) should admit a factorized form obtained by multiplying the cyclic factors over n in [equation \(10\)](#).

Problem 14.2 (Spectral cyclic dilogarithm). *Find a convergent regularized product $\mathcal{D}_\omega(z)$ such that*

$$e^{\mathcal{B}_\omega(z)} = \mathcal{E}_\omega(z) \mathcal{D}_\omega(z),$$

where $\mathcal{E}_\omega$ is an elementary square-root factor and $\mathcal{D}_\omega$ is an infinite spectral product of finite cyclic quantum dilogarithms. Determine its functional equations and zero divisor.

14.3 Irrational boundary phases and small divisors

At an irrational point $q = e^{2\pi i\alpha}$ there is no exact resonance, but denominators $1 - q^{2k}$ can be arbitrarily small. Diophantine properties of α should govern the boundary growth.

Conjecture 14.3 (Diophantine phase transition). *For badly approximable α , suitably Abel-regularized logarithms of $\Phi(re^{2\pi i\alpha}, z)$ admit polynomial bounds as $r \uparrow 1$ on a small frequency disk. For Liouville α , there exist frequency values with superpolynomial subsequential growth generated by near-resonant cumulants.*

This would interpolate between the exact root-of-unity action and generic natural-boundary behavior through small-divisor analysis.

14.4 Arithmetic of the cyclotomic constants

The constants

$$A_\omega = \frac{\zeta(2d)}{2d^2\pi^{2d}}(1 - \omega)^{2d}$$

are algebraic because $\zeta(2d)/\pi^{2d}$ is rational. Their signs are elementary, but their Galois traces, norms, and divisibility encode nontrivial cyclotomic arithmetic.

Problem 14.4 (Galois packets). *For fixed m , determine the trace and norm of A_ω over all primitive m th roots, and similarly for the correction coefficients $C_{k,\omega}$. Relate their prime valuations to cyclotomic units and Bernoulli numerators.*

14.5 Uniform double scaling in root order

The present analysis fixes m . When $m \rightarrow \infty$, the resonance order d grows, A_ω becomes very small, and the limit exponent has increasing degree. There should be a second asymptotic regime balancing d , ε , and w .

Problem 14.5 (Large-order cyclotomic saddle). *Determine a scaling in which $A_\omega w^{2d}$ remains nontrivial as $d \rightarrow \infty$, and identify whether the limiting profile is a hard cutoff, an extreme-value law, or a new Lambert- W saddle. Track the transition between polyharmonic kernels and the endpoint asymptotics already developed in the repository.*

14.6 Inverse monodromy and branch braiding

The formal inverse-frequency series have $2d$ algebraic choices on each logarithm sheet. If sectorial analytic realizations exist, continuation may braid them around critical values of the frequency map, while winding the target around zero changes the logarithm sheet.

Problem 14.6 (Cyclotomic inverse monodromy). *First construct sectorial analytic realizations of the formal inverse branches. Then determine their continuation around simple critical values, where inverse branches may transpose, and separately track the logarithm-sheet shift $\ell \mapsto \ell + 2\pi i$ caused by winding around $y = 0$. The rotation $w_0 \mapsto e^{\pi i/d} w_0$ permutes the $2d$ roots on a fixed nonzero logarithm sheet; it should not be identified with the sheet shift without an additional continuation argument. Classify any resulting braid action and its compatibility with the escaping zero cloud.*

14.7 Two-nome extension

The repository's open problem asks for a two-nome phase diagram and canonical root-of-unity limits. The scalar theorem suggests the correct first invariant: identify the lowest Taylor monomial whose bidegree resonates with the pair of roots. The universal germ calculation suggests studying

$$\prod_{j \geq 0} h((1 - \rho)\rho^j z, (1 - \sigma)\sigma^j w),$$

but the resonance set is now a lattice rather than a single arithmetic progression. Its Newton polygon should determine the anisotropic scaling and the limiting exponential polynomial.

Conjecture 14.7 (Newton-polygon resonance principle). *For an analytic two-variable germ, the canonical two-nome root-of-unity blow-up is determined by the lower convex hull of resonant Taylor exponents. A single vertex gives a monomial exponential; an edge gives a genuinely multivariate exponential polynomial and a coupled polyharmonic PDE.*

14.8 Formalization roadmap

The new theory has a favorable dependency graph. The product layer is already represented by `Fabius.geometricSincProduct_eq_tprod_complexQPochhammerInf`, while the general root-of-unity filter is available as `Fabius.sum_pow_eq_ite` and `Fabius.sum_pow_mul_inv_pow_eq_ite`. Thus the remaining roadmap begins after these reusable foundations.

1. **Finite algebra.** Formalize $d = \text{ord}_{\text{mult}}(\omega^2)$, the resonant/nonresonant split, and the exact first two coefficients of [equation \(15\)](#).
2. **Formal power series.** Prove the coefficient identity [equation \(39\)](#) in a Laurent-series ring; Bell exponentiation is already well suited to existing combinatorial infrastructure.
3. **Finite cyclic projection.** Specialize the existing root-of-unity filter to $d^{-1} \sum_{r < d} \omega^{2kr} = \mathbf{1}_{d|k}$ and derive a finite-truncation version of [equation \(23\)](#).

4. **Analytic tails.** Formalize the square-root cutoff proof of [theorem 4.3](#), including the finite phase separation and exponential tail bound.
5. **Sign and natural boundary.** Formalize the sign of a_ω^{2d} and $\text{Li}_2(\pm x)$ on $0 < x < 1$, then use the finite-order zero/pole contradiction.
6. **Moments.** Reuse the repository’s Bernoulli cumulant and Bell moment APIs to prove the one-cumulant limit.
7. **Inverse branches.** Begin with formal coefficient recursion; defer the analytic implicit-function theorem until the local complex-analysis library is convenient.

A compact first formal milestone would prove the blow-up limit and sparse moment formula without dilogarithms. The natural-boundary theorem can then be added after both the analytic tail estimate and the sign representation are available in Lean.

15 Conclusion

The complex q -plane reveals a structure invisible on the positive real interval. At every root of unity, the geometric sinc product organizes itself by the order of ω^2 . Fixed frequencies see a spectral dilogarithmic action of size $1/\delta$; shrinking frequencies at the exact cyclotomic rate isolate one resonant coefficient and produce a pure exponential $e^{-A_\omega \omega^{2d}}$. The same resonance appears simultaneously in the zero divisor, Bernoulli cumulants, Bell moments, and inverse Appell polynomials.

This organizes several themes of the Fabius–Rvachev corpus while explicitly reusing its existing product, root-of-unity-filter, Bell, Appell, and dyadic infrastructure. The sinc product supplies the spectral side; q -Pochhammer factorization explains the dilogarithmic action; Bell and Bernoulli polynomials control formal corrections and moments; and root order predicts a polyharmonic limit symbol. The two-range estimate below completes the radial theorem and hence the natural-boundary layer. Constructing sectorial inverse branches remains a separate analytic problem.

A Uniform estimates behind the radial theorem

Let $K \subseteq \{z : |a_\omega z| < \pi\}$ and choose $\rho < 1$ such that $|a_\omega z| \leq \rho\pi$ on K . The coefficient bound

$$c_k |a_\omega z|^{2k} \leq \frac{\zeta(2)}{k} \rho^{2k} \quad (84)$$

provides the basic summable majorant. When $d > 1$, the nonresonant phases ω^{2k} range over a finite set not containing one. More explicitly, with

$$\gamma = \min_{\substack{1 \leq r < d \\ 0 \leq t \leq 1}} |1 - \omega^{2r} t| > 0,$$

both denominators occurring in the derivative of $S_k(\delta)$ are bounded below by γ . The numerator derivative and the derivative of $(1 - \delta)^{2k}$ each contribute at most Ck times a geometric majorant. This is the detailed justification of [equation \(20\)](#); when $d = 1$ the nonresonant part is empty.

For resonant k , the cutoff $N = \lfloor \delta^{-1/2} \rfloor$ is deliberately smaller than the tempting scale δ^{-1} . On $k \leq N$ the true expansion parameter $k\delta$ tends uniformly to zero, so Taylor’s theorem with remainder gives [equation \(21\)](#). On $k > N$, geometric decay wins against every power of δ : for each $M > 0$ and $0 < r < 1$,

$$\delta^{-M} r^{\delta^{-1/2}} \longrightarrow 0. \quad (85)$$

This proves, rather than merely predicts, the uniform $O_K(\delta)$ remainder in [theorem 4.3](#).

The same cutoff method can be repeated to any prescribed finite order, but a full Poincaré theorem requires keeping the order-dependent Taylor constants explicit. This suggests the further expansion

$$\log \Phi(\omega(1 - \delta), z) \sim -\frac{\mathcal{A}_\omega(z)}{\delta} + \sum_{r \geq 0} B_{r,\omega}(z) \delta^r \quad (86)$$

locally uniformly in the same disk. Each candidate $B_{r,\omega}$ is an explicitly convergent zeta-weighted coefficient series with rational functions of k and ω as coefficients. We leave this all-orders fixed-frequency statement as a conjectural strengthening; it is not needed for any theorem in the main text.

B A direct coefficient derivation for the inverse q -branch

Set $\delta = \rho z^{2d}$. The $k = d$ term in the logarithm is

$$-c_d z^{2d} \frac{(a_\omega + \omega \rho z^{2d})^{2d}}{1 - (1 - \rho z^{2d})^{2d}} = -\frac{A_\omega}{\rho} + O(z^{2d}). \quad (87)$$

For $1 \leq k < d$, the nonresonant terms are

$$-C_{k,\omega} z^{2k} + O(z^{2k+2d}). \quad (88)$$

Write

$$\rho(z) = \rho_0 + \rho_1 z^2 + O(z^4), \quad \rho_0 = -\frac{A_\omega}{\ell}. \quad (89)$$

Then

$$-\frac{A_\omega}{\rho(z)} = \ell + \ell \left(-\frac{\rho_1}{\rho_0} \right) z^2 + O(z^4).$$

Matching the $-C_{1,\omega} z^2$ term gives $\rho_1/\rho_0 = -C_{1,\omega}/\ell$, which is [equation \(82\)](#). Higher coefficients follow triangularly because the derivative in ρ never vanishes at ρ_0 .

C Exact small-order constants

Table 2: Leading constants for selected primitive roots $\omega = e^{2\pi i/m}$.

m	d	a_ω^{2d}	A_ω	real-axis type
2	1	4	1/3	dissipative Gaussian
3	3	-27	-1/630	backward sixth order
4	2	-4	-1/180	backward fourth order
6	3	1	1/17010	dissipative sixth order

For general primitive order m , the sign rule is

$$A_\omega > 0 \iff m \equiv 2 \pmod{4}, \quad A_\omega < 0 \iff m \text{ is odd or } 4 \mid m.$$

The magnitude can be written

$$|A_\omega| = \frac{\zeta(2d)}{2d^2 \pi^{2d}} \left(2 \sin \frac{\pi h}{m} \right)^{2d}. \quad (90)$$

D Reproducibility manifest

The companion archive contains:

path	purpose
cyclotomic_q_fabius_frontier.tex	complete manuscript
cyclotomic_q_fabius_frontier.pdf	compiled and visually inspected PDF
numerical_experiments.py	commented high-precision experiments
data/*.csv	raw deterministic numerical tables
data/experiment_summary.txt	concise diagnostic output
figures/*.pdf, figures/*.png	vector and raster figures
corpus_audit_note.txt	audit method and novelty boundary
corpus_inventory_2026-08-27.txt	preserved 57-path recursive ledger
requirements.txt	Python dependencies
README.txt	build and reproduction instructions

The script uses the exact convergent logarithmic series rather than a truncated product. To reproduce the tables and figures, run

```
python numerical_experiments.py --dps 80
```

from the archive root. The LaTeX source expects the generated figure and data directories in their packaged relative locations.

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